MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 17: Tuesday 13 October 2026

(Week 9)

Remark. Q15.8 is a leftover exercise from Week 8.

Remark. Q15.9 is supplementary exercise on double integrals (Weel 8).

Remark. Q17.1 to Q17.4 are on Section 14.8 of Thomas' Calculus.

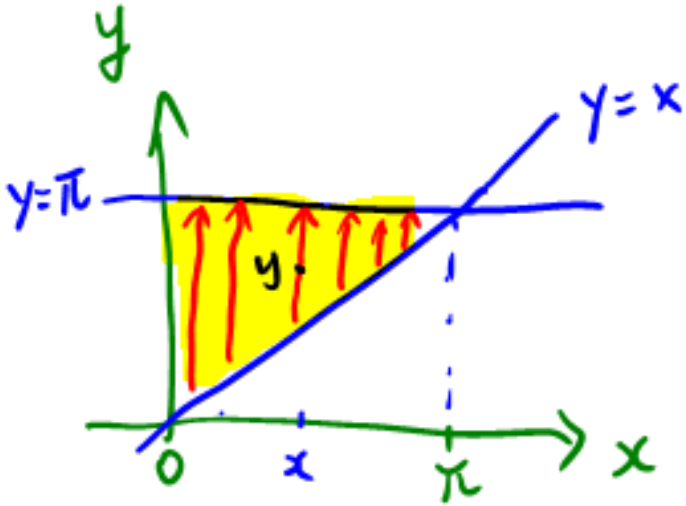
Q15.8. Evaluate the following integral by reversing the order of integration.

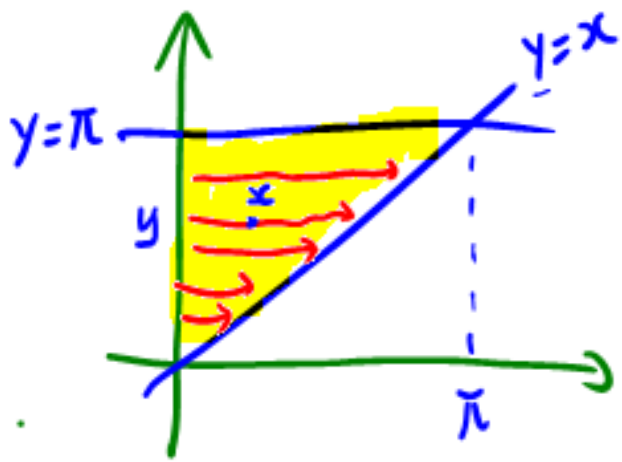
$$\int_0^{\pi}\left(\int_x^{\pi}\frac{\sin y}{y}\,dy\right)dx. = \int_{y=0}^{y=\pi}\int_{x=0}^{x=y}\frac{\sin y}{y}\,dx\,dy$$

$R = \{(x,y): 0 \le x \le \pi,\ x \le y \le \pi\}$

$= \{(x,y): 0 \le y \le \pi,\ 0 \le x \le y\}$

$\int \frac{\sin y}{y}\,dy = ??$





tricky variable: x

simple variable: y

Solution:

$$\int_0^{\pi} \int_x^{\pi} \frac{\sin y}{y}\, dy\, dx = \iint_R \frac{\sin y}{y}\, dA,$$

where R is the region

$$R = \{(x, y) \,:\, 0 \leq x \leq \pi, \;\; x \leq y \leq \pi\}.$$

We rewrite R as:

$$R = \{(x, y) : 0 \le y \le \pi, \ \ 0 \le x \le y\}.$$

We have

$$\int_0^\pi \int_x^\pi \frac{\sin y}{y}\, dy\, dx$$

$$= \iint_R \frac{\sin y}{y}\, dA$$

$$= \int_0^\pi \left(\int_0^y \frac{\sin y}{y}\, dx \right) dy$$

$$= \int_0^\pi \left[\frac{\sin y}{y} \cdot x \right]_0^y dy \quad = \lim_{a \to 0+} \int_a^\pi \left[\frac{\sin y}{y} \cdot x \right]_0^y dy$$

$$= \int_0^\pi \frac{\sin y}{y} \cdot (y - 0)\, dy$$

$$= \int_0^\pi \sin y\, dy = [-\cos y]_0^\pi = -(-1) - (-1) = 2.$$

More formally, we have

$$\int_0^\pi \int_x^\pi \frac{\sin y}{y}\, dy\, dx$$

$$= \iint_R \frac{\sin y}{y}\, dA$$

$$= \int_0^\pi \int_0^y \frac{\sin y}{y}\, dx\, dy$$

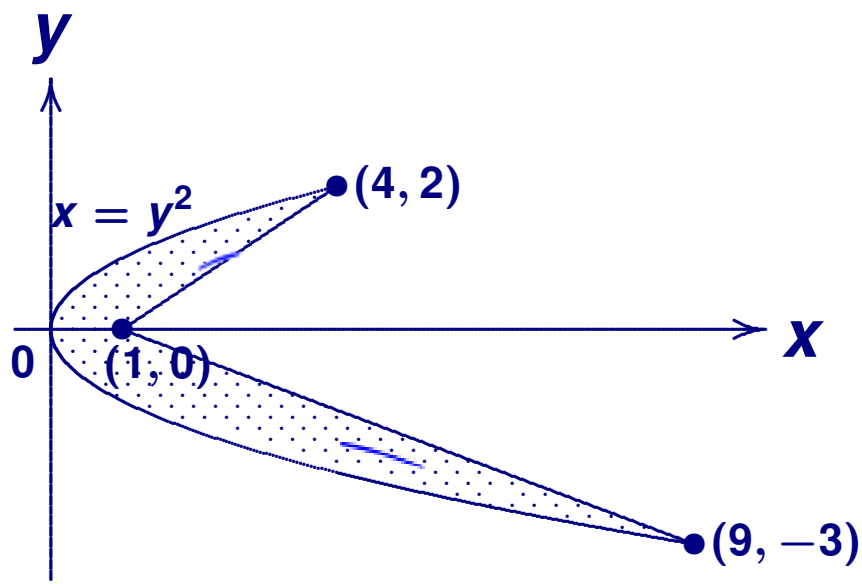
$$= \lim_{a \to 0+} \int_a^\pi \left[\frac{\sin y}{y} \cdot x \right]_0^y dy$$

$$= \lim_{a \to 0+} \int_a^\pi \frac{\sin y}{y} \cdot (y - 0)\, dy$$

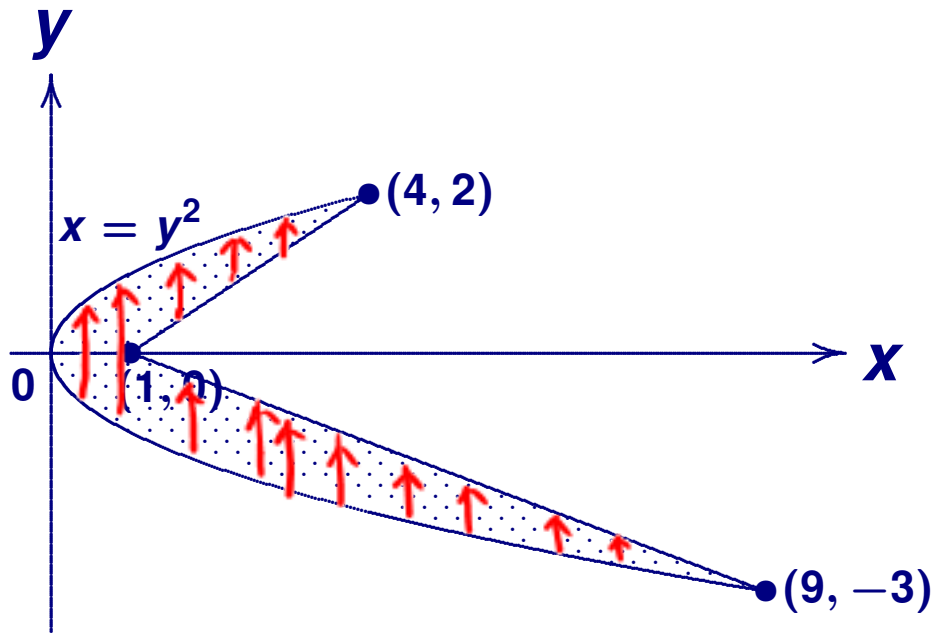
$$= \lim_{a \to 0+} \int_a^\pi \sin y\, dy = \lim_{a \to 0+} \Big[-\cos y \Big]_a^\pi$$

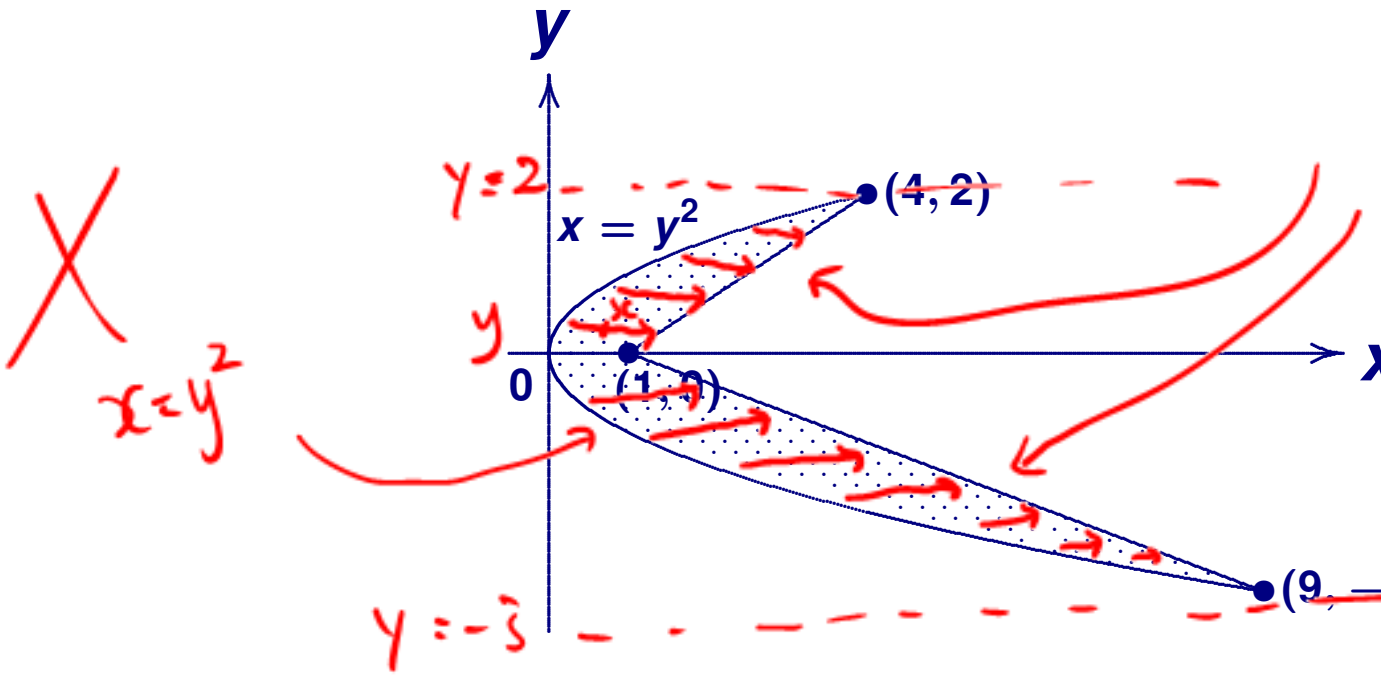
$$= \lim_{a \to 0+} \Big(-(-1) - (-\cos a) \Big) = 1 + 1 = 2.$$

Q15.9. (Supp. Exercise) Evaluate the double integral $\iint_D 10x\,dA$ where D is the shaded region as shown below. (The boundary of D consists of the line segment joining the two points $(1, 0)$ and $(4, 2)$, the line segment joining the points $(1, 0)$ and $(9, -3)$, and the part of the parabola $x = y^2$ joining the points $(9, -3)$ and $(4, 2)$.)



[Suggestion: Compute the double integral as a sum of two double integrals over subregions.]



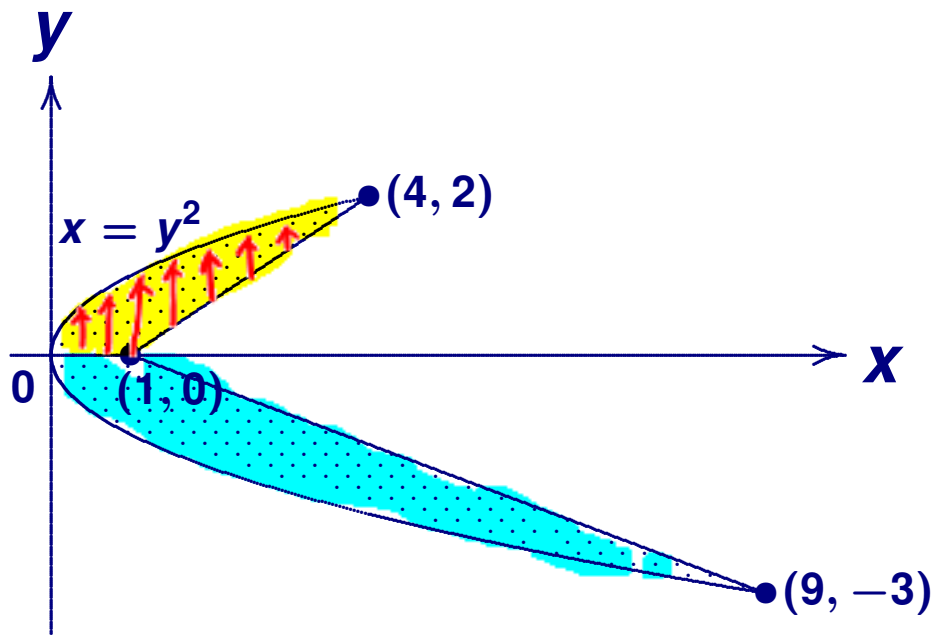


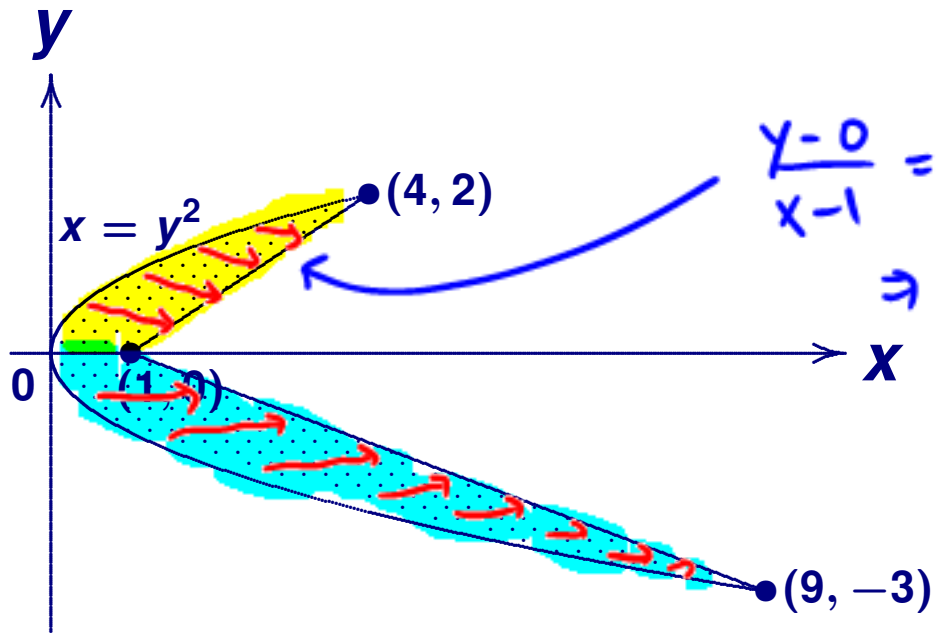
X

X

$x = y^2$

The bounding curve on the right consists of 2 different line segments defined by different equations!





$$\frac{y-0}{x-1} = \frac{2-0}{4-1} = \frac{2}{3}$$

$$\Rightarrow \quad y = \frac{2}{3}x - \frac{2}{3}$$

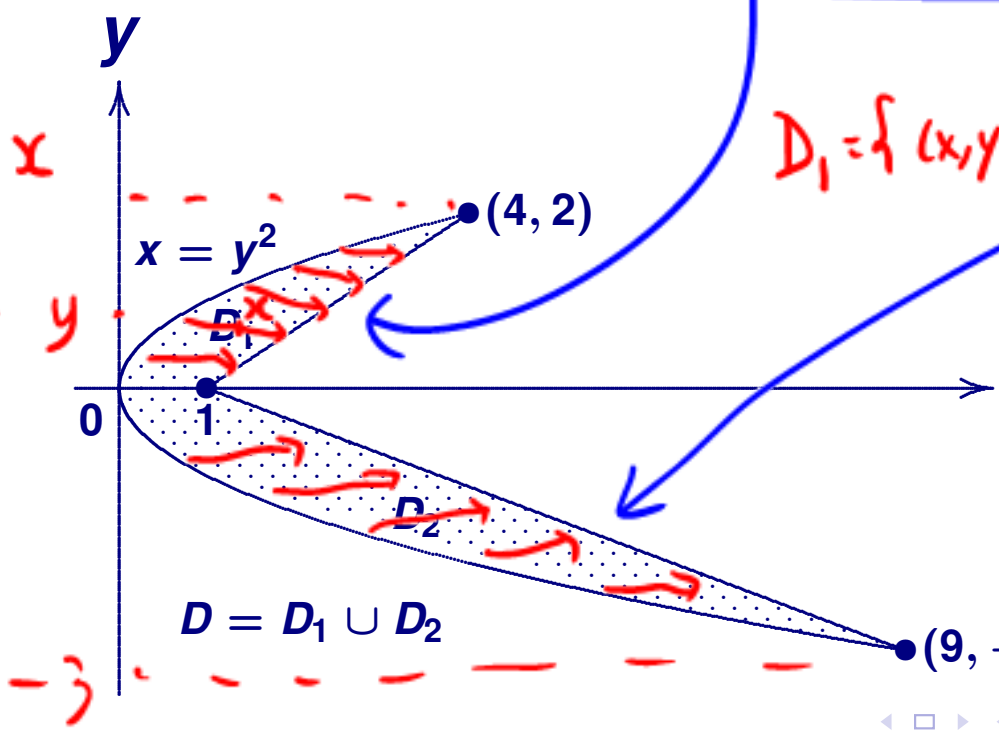
$$\Rightarrow \quad x = \frac{3}{2}y + 1$$

Solution: The line joining $(1, 0)$ and $(4, 2)$ has equation

$$\frac{y-0}{x-1} = \frac{2-0}{4-1} = \frac{2}{3} \Leftrightarrow y = \frac{2}{3}x - \frac{2}{3} \Leftrightarrow x = \frac{3}{2}y + 1.$$

The line joining $(1, 0)$ and $(9, -3)$ has equation

$$\frac{y-0}{x-1} = \frac{(-3)-0}{9-1} = -\frac{3}{8} \Leftrightarrow y = -\frac{3}{8}x + \frac{3}{8} \Leftrightarrow x = -\frac{8}{3}y + 1.$$



Tricky variable: x

simple variable: y

$D_1 = \{(x, y) : 0 \le y \le 2,$

$y^2 \le x \le \frac{3}{2}y + 1\}$

$$\int_{y=0}^{y=2} \int_{x=y^2}^{x=\frac{3}{2}y+1} 10x \, dx \, dy$$

The region D is the union of D_1 and D_2, where

$$D_1 = \{(x, y) \in \mathbb{R}^2 \mid y^2 \le x \le \frac{3}{2}y + 1, \quad 0 \le y \le 2\},$$

$$D_2 = \{(x, y) \in \mathbb{R}^2 \mid y^2 \le x \le -\frac{8}{3}y + 1, \quad -3 \le y \le 0\}.$$

Hence the required answer is

$$\iint_D 10x \, dA$$

$$= \iint_{D_1} 10x \, dA + \iint_{D_2} 10x \, dA$$

$$= \int_0^2 \left(\int_{y^2}^{(3y/2)+1} 10x \, dx \right) dy + \int_{-3}^{0} \int_{y^2}^{-(8y/3)+1} 10x \, dx \, dy$$

$$= 38 + 212 = 250,$$

since

$$\int_0^2 \int_{y^2}^{(3y/2)+1} 10x \, dxdy$$

$$= \int_0^2 \left[5x^2\right]_{x=y^2}^{x=(3y/2)+1} dy$$

$$= \int_0^2 5\left(\frac{9y^2}{4} + 3y + 1 - y^4\right) dy$$

$$= \left[\frac{15y^3}{4} + \frac{15y^2}{2} + 5y - y^5\right]_0^2$$

$$= (30 + 30 + 10 - 32) - 0 = 38,$$

and

$$\int_{-3}^{0}\int_{y^2}^{-(8y/3)+1} 10x\,dxdy = \int_{-3}^{0}\left[5x^2\right]_{x=y^2}^{x=-(8y/3)+1} dy$$

$$= \int_{-3}^{0} 5\left(\frac{64y^2}{9} - \frac{16y}{3} + 1 - y^4\right) dy$$

$$= \left[\frac{320y^3}{27} - \frac{40y^2}{3} + 5y - y^5\right]_{-3}^{0}$$

$$= 0 - (-320 - 120 - 15 + 243)$$

$$= 212.$$

Change of variables (substitution) in double integrals

Recall integration by substitution:

Suppose we want to compute $\int_a^b f(x)\,dx$ with $a < b$,

and suppose New variable: u

$x = a \longleftrightarrow u = c$

$x = b \longleftrightarrow u = d$

$$x = x(u), \quad x(c) = a, \quad x(d) = b.$$

Then $$\int_a^b f(x)\,dx = \int_c^d f(x(u))\,\frac{dx}{du}\,du. \qquad (1)$$

Note that

$$\frac{dx}{du} = \frac{1}{\dfrac{du}{dx}} = \left(\frac{du}{dx}\right)^{-1}.$$

So we may also write (1) as

$$\int_a^b f(x)\,dx = \int_c^d f(x(u))\left(\frac{du}{dx}\right)^{-1} du. \qquad (2)$$

If $\dfrac{dx}{du} < 0$, then

$$a < b \implies d < c, \quad \text{and} \quad \frac{dx}{du} = -\left|\frac{dx}{du}\right|.$$

So if we want to have lower limit $<$ upper limit, we may write (1) as

$$\int_a^b f(x)\,dx = \int_d^c f(x(u))\left|\frac{dx}{du}\right| du. \qquad (3)$$

Simiarly, (2) becomes

$$\int_a^b f(x)\,dx = \int_d^c f(x(u))\left|\frac{du}{dx}\right|^{-1} du. \qquad (4)$$

Let R be a region in the xy-plane. Suppose we want to compute $= \iint_R f(x, y)\, dA = \iint_R f(x, y)\, dx\, dy$.

Suppose a transformation / change of variables

new variables: u, v

$$x = x(u, v), \quad y = y(u, v) \qquad (5)$$

sends a region G in the uv-plane to the region R in the xy-plane in a one-to-one manner.

Denote the **Jacobian determinant** of the transformation in (5) by

$$\frac{\partial(x, y)}{\partial(u, v)} \overset{\begin{matrix} x \\ y \end{matrix}}{=} \begin{matrix} u & v \\ \end{matrix} \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \leftarrow \text{determinant} = \frac{\partial x}{\partial u} \cdot \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \cdot \frac{\partial y}{\partial u}.$$

Then as in (3), we have

G corresponding region in u, v plane.

$$\iint_R f(x,y)\,dx\,dy = \iint_G f(x(u,v),y(u,v))\left|\frac{\partial(x,y)}{\partial(u,v)}\right|\,du\,dv.$$

absolute value

Similar to (2), we may write (4) as

$$\iint_R f(x,y)\,dx\,dy = \iint_G f(x(u,v),y(u,v))\left|\frac{\partial(u,v)}{\partial(x,y)}\right|^{-1}\,du\,dv.$$

An informal derivation of the change of variable formula:

Suppose we have a small rectangle in the uv-plane spanned by the two vectors $\langle \triangle u, 0 \rangle$ and $\langle 0, \triangle v \rangle$ with tails of both vectors at a point (u, v). Suppose the transformation sends the point (u, v) to a point (x, y) in the xy-plane. Then the transformation will send the vector $\langle \triangle u, 0 \rangle$ in the uv-plane approximately to the vector $\langle \frac{\partial x}{\partial u} \triangle u, \frac{\partial y}{\partial u} \triangle u \rangle$ in the xy-plane. Similarly, the transformation will send the vector $\langle 0, \triangle v \rangle$ approximately to the vector $\langle \frac{\partial x}{\partial v} \triangle v, \frac{\partial y}{\partial v} \triangle v \rangle$.

Then the small rectangle of area $\triangle A_{u,v} = \triangle u \cdot \triangle v$ in the uv-plane will be mapped to a parallelogram in the xy-plane with area ($\triangle A_{x,y}$) equal to

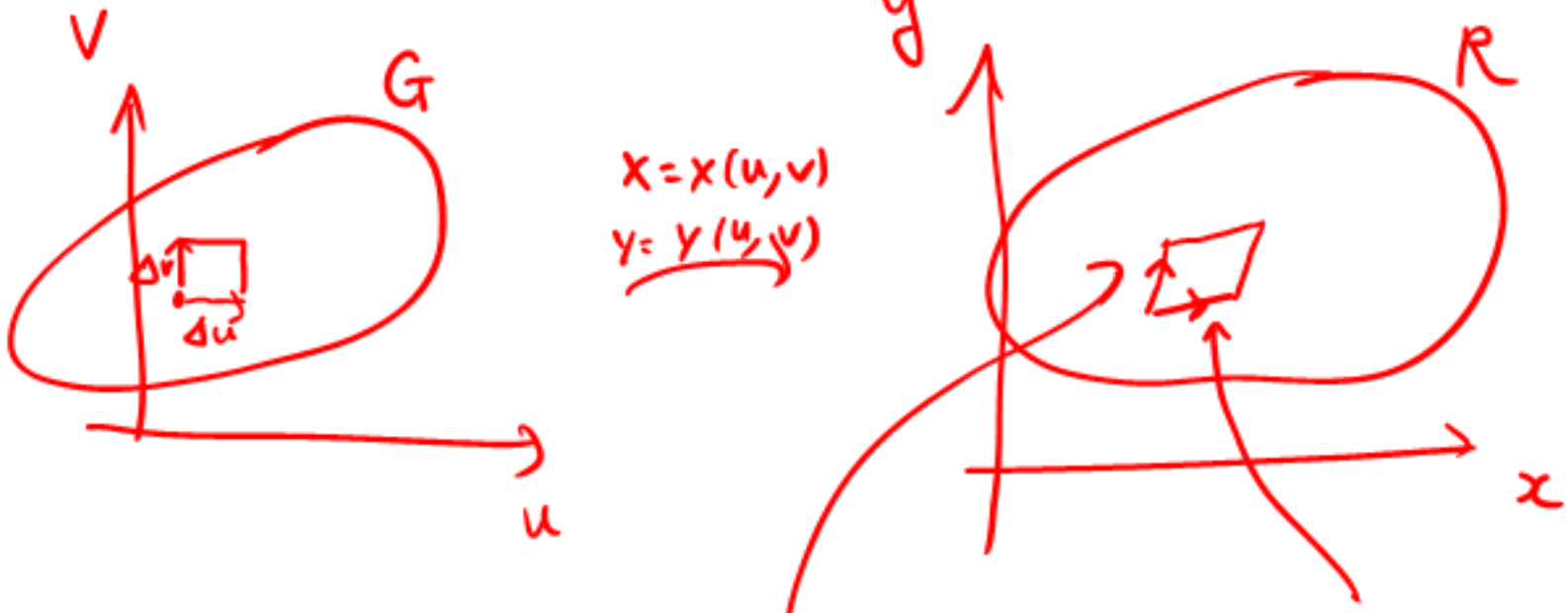
$$\triangle A_{x,y} \approx \left\| \begin{vmatrix} \frac{\partial x}{\partial u}\triangle u & \frac{\partial y}{\partial u}\triangle u \\ \frac{\partial x}{\partial v}\triangle v & \frac{\partial y}{\partial v}\triangle v \end{vmatrix} \right\| = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| \triangle u \triangle v.$$

Now letting $\triangle x, \triangle y \to 0$, we get

$$dA_{x,y} = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| dA_{u,v},$$

or equivalently,

$$dx\,dy = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du\,dv.$$



$$\langle \Delta u, 0 \rangle \longrightarrow \left\langle \frac{\partial x}{\partial u}\Delta u, \frac{\partial y}{\partial u}\cdot \Delta u \right\rangle$$

$$\langle 0, \Delta v \rangle \longrightarrow \left\langle \frac{\partial x}{\partial v}\Delta v, \frac{\partial y}{\partial v}, \Delta v \right\rangle$$

$$dA_{u,v} = \Delta u \cdot \Delta v, \longleftrightarrow dA_{x,y} = \left| \left| \begin{matrix} \frac{\partial x}{\partial u}\Delta u & \frac{\partial y}{\partial u}\Delta u \\ \frac{\partial x}{\partial v}\Delta v & \frac{\partial y}{\partial v}\Delta v \end{matrix} \right| \right|$$

$$= \left| \frac{\partial(x,y)}{\partial(u,v)} \right| \cdot \Delta u \Delta v$$

Q17.1. Consider the transformation from the uv-plane to the xy-plane is given by the equations:

$$x = \frac{u - v}{\sqrt{2}}, \quad y = \frac{u + v}{\sqrt{2}}.$$

Suppose the image of a region G in the uv-plane under the above transformation is the region $R = \{(x, y) : 0 \leq x \leq 1, \ \ 0 \leq y \leq 1\}$ in the xy-plane. Find G.

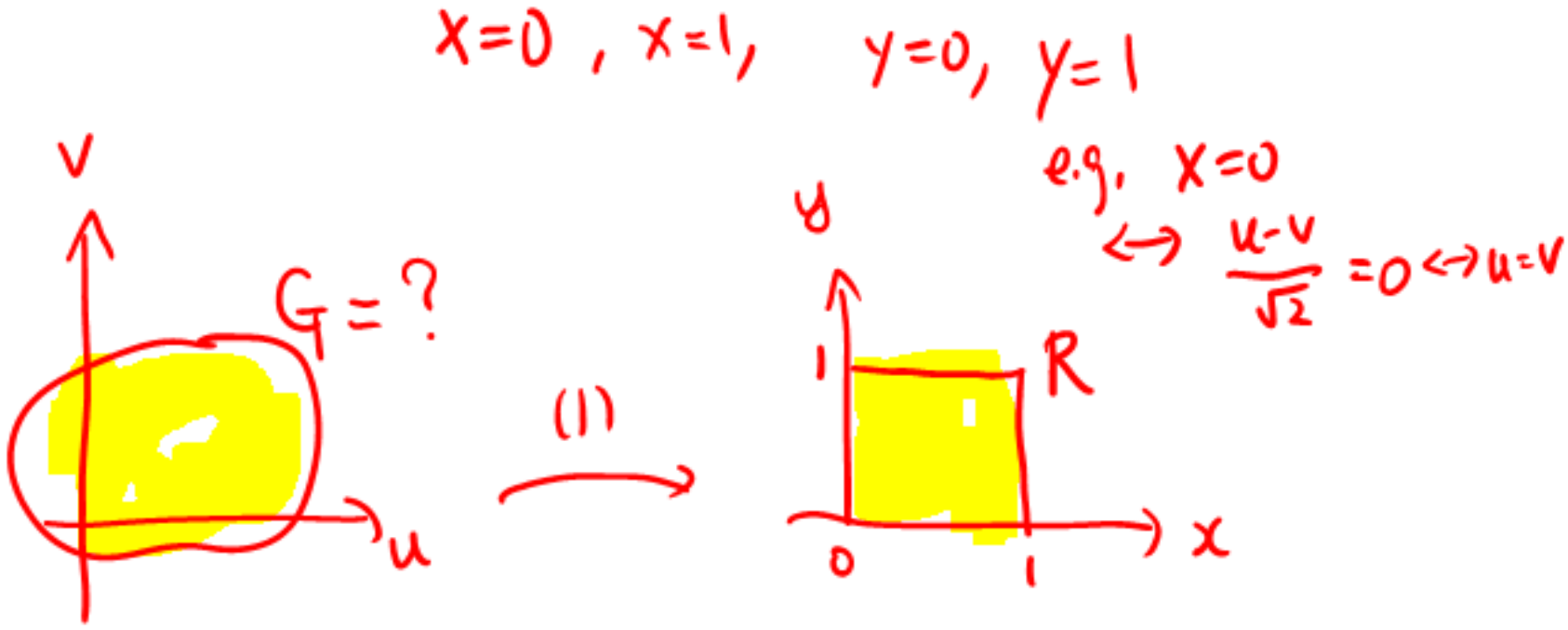
[Remark: First determine the boundaries (boundary lines) of the region R. Then get the corresponding lines in the uv-plane. Then describe G by inequalities involving u and v.]

Solution: Given the transformation from the uv-plane to the xy-plane given by the equations:

$$x = \frac{u - v}{\sqrt{2}}, \quad y = \frac{u + v}{\sqrt{2}}. \qquad (1)$$

Consider the set in the xy-plane given by

$$R = \{(x, y) : 0 \leq x \leq 1, \ \ 0 \leq y \leq 1\}.$$



Observe that x and y are linear polynomials in u and v. Thus the transformation will send lines in uv-planes to lines in xy-plane.

The boundaries of the region R in the xy-planes are

$$x = 0, \quad x = 1, \quad y = 0, \quad y = 1.$$

For the line $x = 0$, note that from (1), we have

$$x = 0 \iff \frac{u - v}{\sqrt{2}} = 0 \iff u = v.$$

Similarly, $x = 1 \iff \dfrac{u - v}{\sqrt{2}} = 1 \iff u = v + \sqrt{2};$

$$y = 0 \iff \frac{u + v}{\sqrt{2}} = 0 \iff u = -v;$$

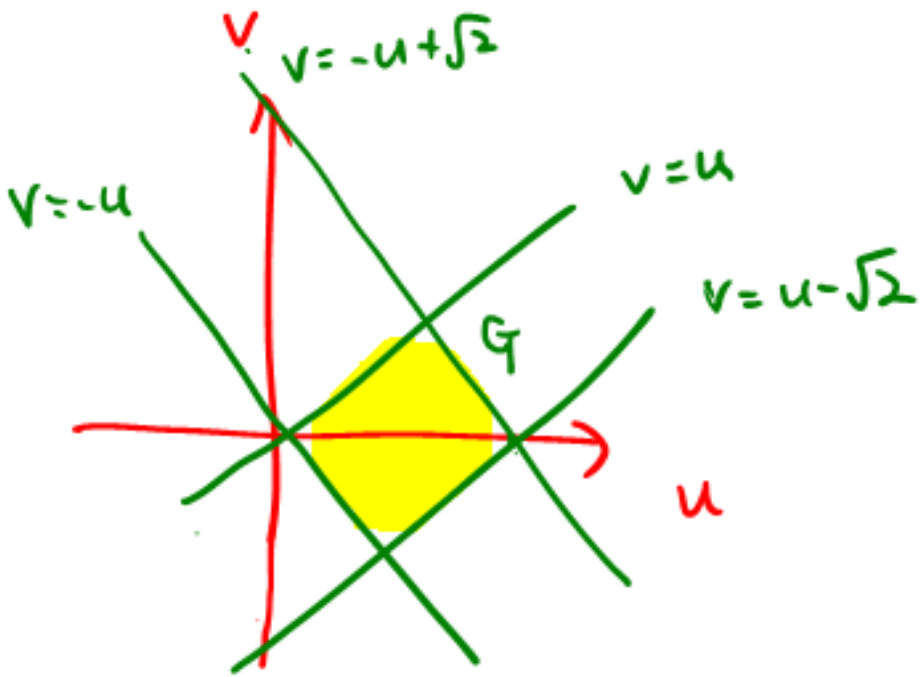
$$y = 1 \iff \frac{u + v}{\sqrt{2}} = 1 \iff u = -v + \sqrt{2}.$$

Then the region G in the uv-plane will be the region bounded by the lines

$$u = v, \quad u = v + \sqrt{2}, \quad u = -v, \quad u = -v + \sqrt{2},$$

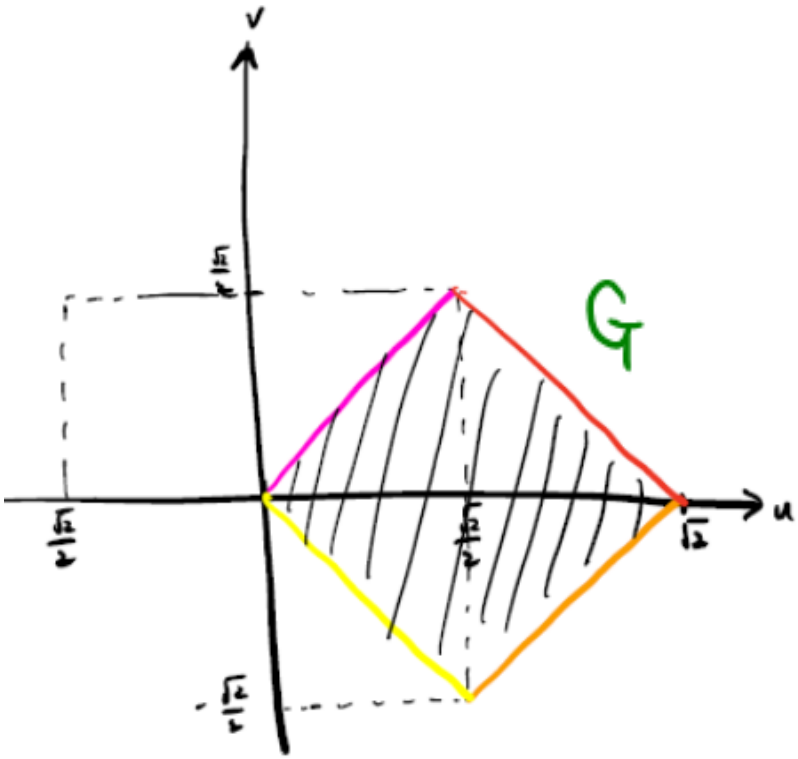
which can be rewritten as

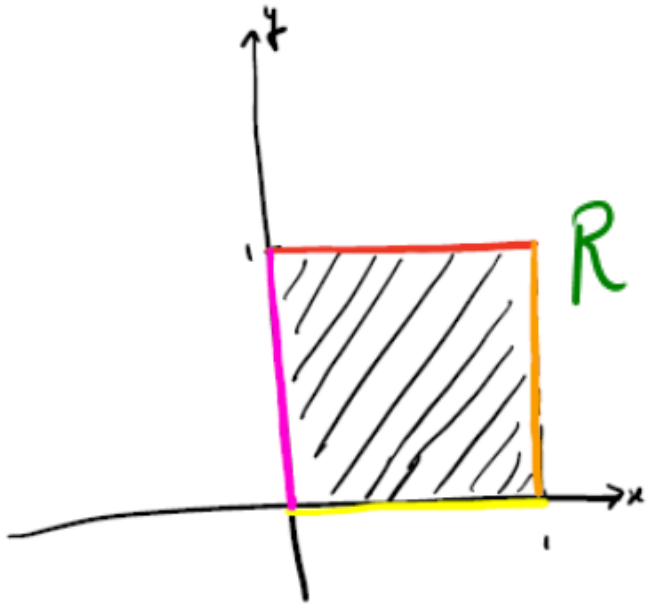
$$v = u, \quad v = u - \sqrt{2}, \quad v = -u, \quad v = -u + \sqrt{2}.$$



Thus the region G is given by:

$$G = \{(u, v) : u - \sqrt{2} \leq v \leq u \ \& \ -u \leq v \leq -u + \sqrt{2}\}.$$

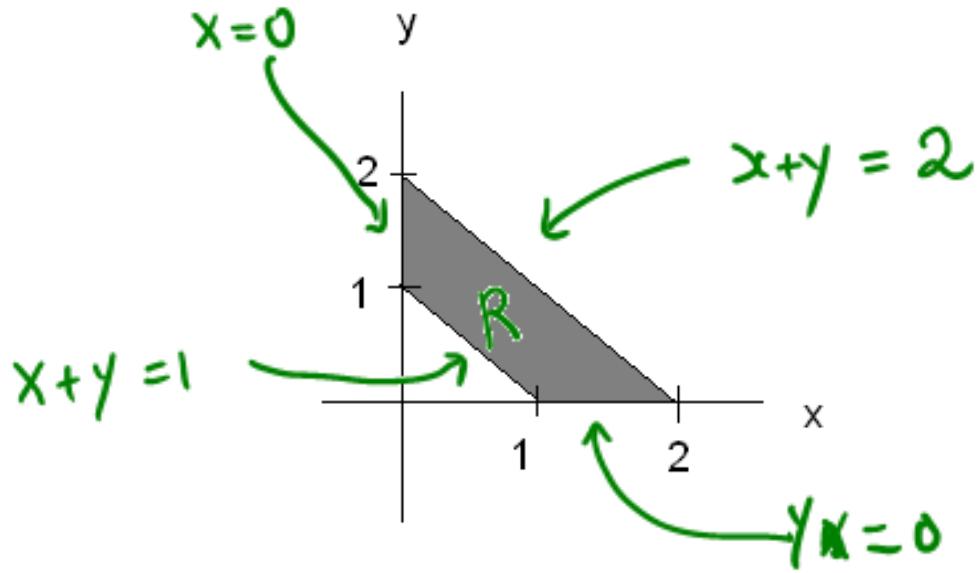




Q17.2. Evaluate

$$\iint_R \cos\left(\frac{y-x}{y+x}\right)\, dA,$$

where R is the trapezoidal region with vertices $(1,0)$, $(2,0)$, $(0,2)$ and $(0,1)$.



Suggestion: (i) Set $u = y - x$ and $v = y + x$.

(ii) First determine the corresponding region G in the uv-plane as in Q17.1.

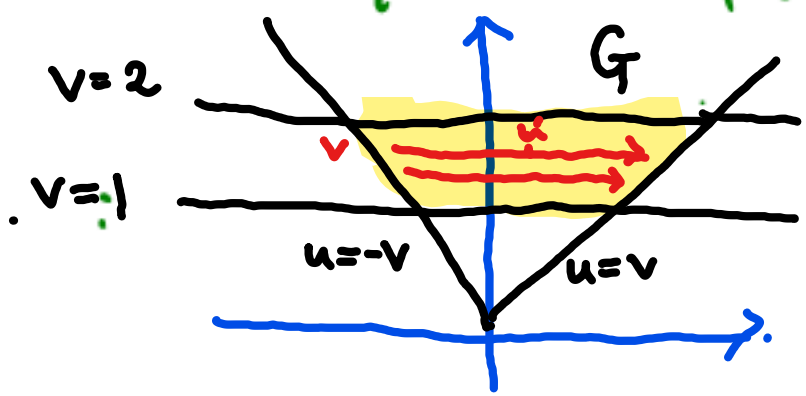
Solution: change of variables: $u = y - x$, $v = y + x$.

R: bounding lines: $x = 0$, $y = 0$, $x + y = 1$, $x + y = 2$.

G: bounding lines: $v = u$ ($u = v$), $u + v = 0$ ($u = -v$), $v = 1$, $v = 2$

$$\begin{cases} u = y - x \\ v = y + x \end{cases} \Leftrightarrow \begin{cases} x = \dfrac{v-u}{2} \\ y = \dfrac{u+v}{2} \end{cases}$$

$$G = \{ (u,v) : \ 1 \le v \le 2, \ -v \le u \le v \}$$



tricky variable: u

simple variable: v

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix}$$

$$= -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}.$$

$x = \frac{v-u}{2}$

$y = \frac{u+v}{2}$

$$\iint_R \cos\left(\frac{y-x}{x+y}\right) dx\,dy \quad \therefore$$

$$= \iint_G \cos\left(\frac{u}{v}\right) \cdot \left|\frac{\partial(x,y)}{\partial(u,v)}\right| \cdot du\,dv$$

$$= \int_{v=1}^{v=2} \int_{u=-v}^{u=v} \cos\left(\frac{u}{v}\right) \cdot \left|-\frac{1}{2}\right| du\,dv$$

$$= \int_{v=1}^{v=2} \left(\int_{u=-v}^{u=v} \cos\frac{u}{v} \cdot \frac{1}{2}\, du \right) dv$$

$$= \int_{v=1}^{v=2} \left[\frac{v}{2} \sin\left(\frac{u}{v}\right) \right]_{u=-v}^{u=v} dv$$

$$= \int_{v=1}^{v=2} \frac{v}{2}\left(\sin 1 - \sin(-1)\right) dv$$

$$= \int_{v=1}^{v=2} v \cdot \sin 1 \, dv$$

$$= \left[\sin 1 \cdot \frac{v^2}{2} \right]_{v=1}^{v=2} = \sin 1 \cdot \left(\frac{2^2}{2} - \frac{1}{2} \right)$$

$$= \frac{3}{2} \sin 1.$$